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**Bisinger**

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(45) **Date of Patent: Aug. 26, 2025**

(54) **COLLAPSIBLE CONTAINERS INCLUDING EDGE ATTACHMENT BRACKETS**

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(51) **Int. Cl.**
***B65D 88/52*** (2006.01)
***B65D 90/08*** (2006.01)

(52) **U.S. Cl.**
CPC ........... ***B65D 90/08*** (2013.01); ***B65D 88/526*** (2013.01)

(58) **Field of Classification Search**
CPC .............................. B65D 90/08; B65D 88/526
USPC .......................................... 206/600; 220/4.33
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,526,113 A * 10/1950 Blackwell ............ B65D 85/305 217/65
3,266,656 A * 8/1966 Kridle ...................... B65D 9/12 220/684
3,477,604 A * 11/1969 Kridle ...................... B65D 9/12 217/12 R
3,989,156 A 11/1976 Lowry
4,024,977 A 5/1977 Rowley
4,796,392 A 1/1989 Graham, Jr.
5,114,194 A 5/1992 Toifl et al.
5,236,099 A * 8/1993 Fties .................. B65D 19/0016 220/4.31
5,253,763 A 10/1993 Kirkley et al.
D347,092 S * 5/1994 Ravel ........................... D30/118
5,429,259 A * 7/1995 Robin ...................... B65D 9/34 217/65
5,555,980 A 9/1996 Johnston et al.
5,566,849 A * 10/1996 Goehner .................. B65D 9/14 220/6

(Continued)

FOREIGN PATENT DOCUMENTS

EP 0455875 B1 3/1995

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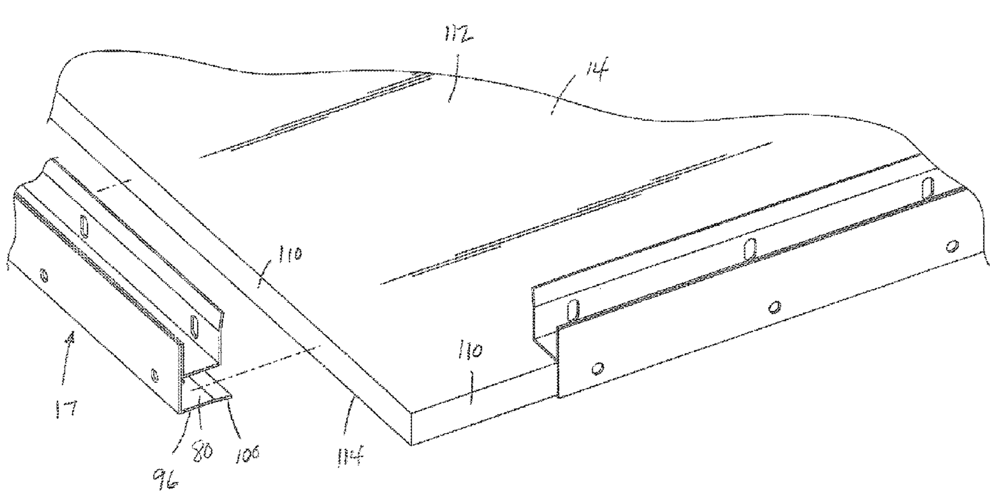
(57) **ABSTRACT**

A container that is formed from a plurality of panels that are joined to each other by a plurality of edge brackets. Each of the edge brackets includes a first channel member and a second channel member that are joined to each other. The first channel member includes a first receiving channel and the second channel member includes a second receiving channel. The first and second receiving channels are arranged to be perpendicular to each other and are each sided to receive a side edge of one of the panels that form the container. The series of edge brackets can be used to securely assembly the container from the plurality of panels after the disassembled container has been shipped.

**16 Claims, 13 Drawing Sheets**



(56) **References Cited**

U.S. PATENT DOCUMENTS

| | | | | |
|---|---|---|---|---|
| 5,597,084 | A | 1/1997 | Parasin | |
| 5,722,551 | A * | 3/1998 | Cocciemiglio, Jr. | B65D 9/24 220/4.33 |
| 6,315,495 | B1 | 11/2001 | Starheim | |
| D460,263 | S * | 7/2002 | Bradford | D3/318 |
| 6,443,508 | B1 | 9/2002 | Persson | |
| 6,554,148 | B1 * | 4/2003 | Fernandez | B65D 11/1873 220/4.33 |
| 6,749,077 | B1 | 6/2004 | McAlpine et al. | |
| 6,799,691 | B2 | 10/2004 | Wincent | |
| 7,428,973 | B2 | 9/2008 | Barth et al. | |
| 7,552,826 | B2 | 6/2009 | Watanabe | |
| 7,591,053 | B2 * | 9/2009 | Bosisio | E04H 13/006 27/35 |
| 7,753,222 | B1 | 7/2010 | Heinrichs | |
| 8,028,849 | B2 | 10/2011 | Hall | |
| 8,083,448 | B2 | 12/2011 | Heinrichs et al. | |
| 8,672,137 | B2 | 3/2014 | Seagle et al. | |
| 8,708,178 | B2 | 4/2014 | Kvols et al. | |
| 8,813,985 | B2 | 8/2014 | Scholesser | |
| 8,915,390 | B2 * | 12/2014 | Meyer-Horn | B65D 9/24 217/43 A |
| 10,060,455 | B2 | 8/2018 | Munch-Fals | |
| 10,858,167 | B2 * | 12/2020 | D'Anglade | B65D 71/0096 |
| 11,731,805 | B2 * | 8/2023 | Bisinger | B65D 19/20 206/600 |
| 2002/0040903 | A1 * | 4/2002 | Coones | B65D 9/12 220/4.33 |
| 2002/0139798 | A1 | 10/2002 | Logan et al. | |
| 2004/0131416 | A1 * | 7/2004 | Wu | A47B 47/0033 403/170 |
| 2004/0208689 | A1 | 10/2004 | Dijkstra | |
| 2006/0226143 | A1 | 10/2006 | Elstone et al. | |
| 2007/0000923 | A1 * | 1/2007 | Chen | B65D 19/06 220/4.33 |
| 2011/0194893 | A1 | 8/2011 | Wiebe | |
| 2013/0315652 | A1 | 11/2013 | Eding | |
| 2014/0110404 | A1 | 4/2014 | Wiebe | |
| 2016/0195118 | A1 | 7/2016 | Munch-Fals et al. | |

* cited by examiner

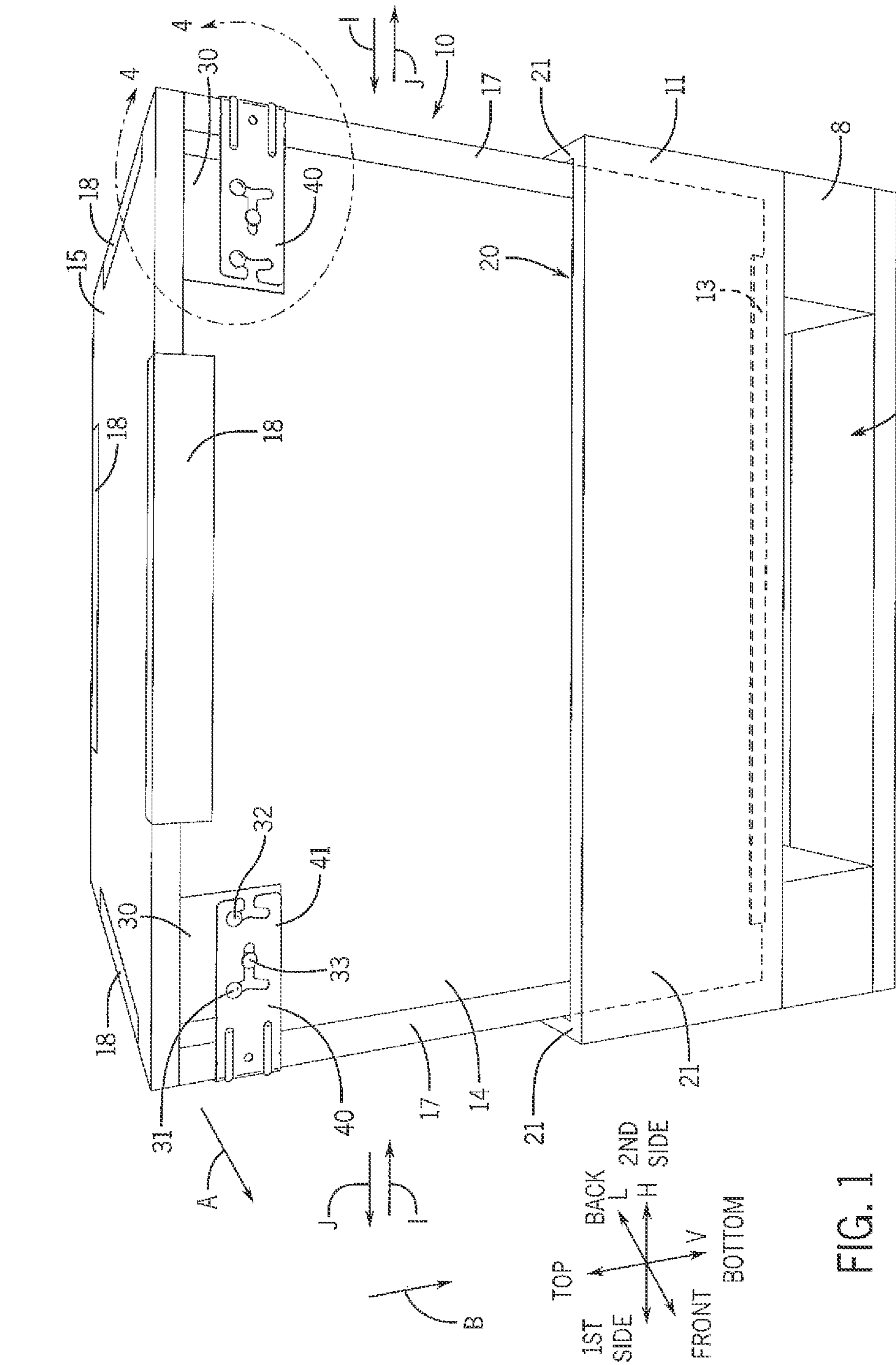



FIG. 1

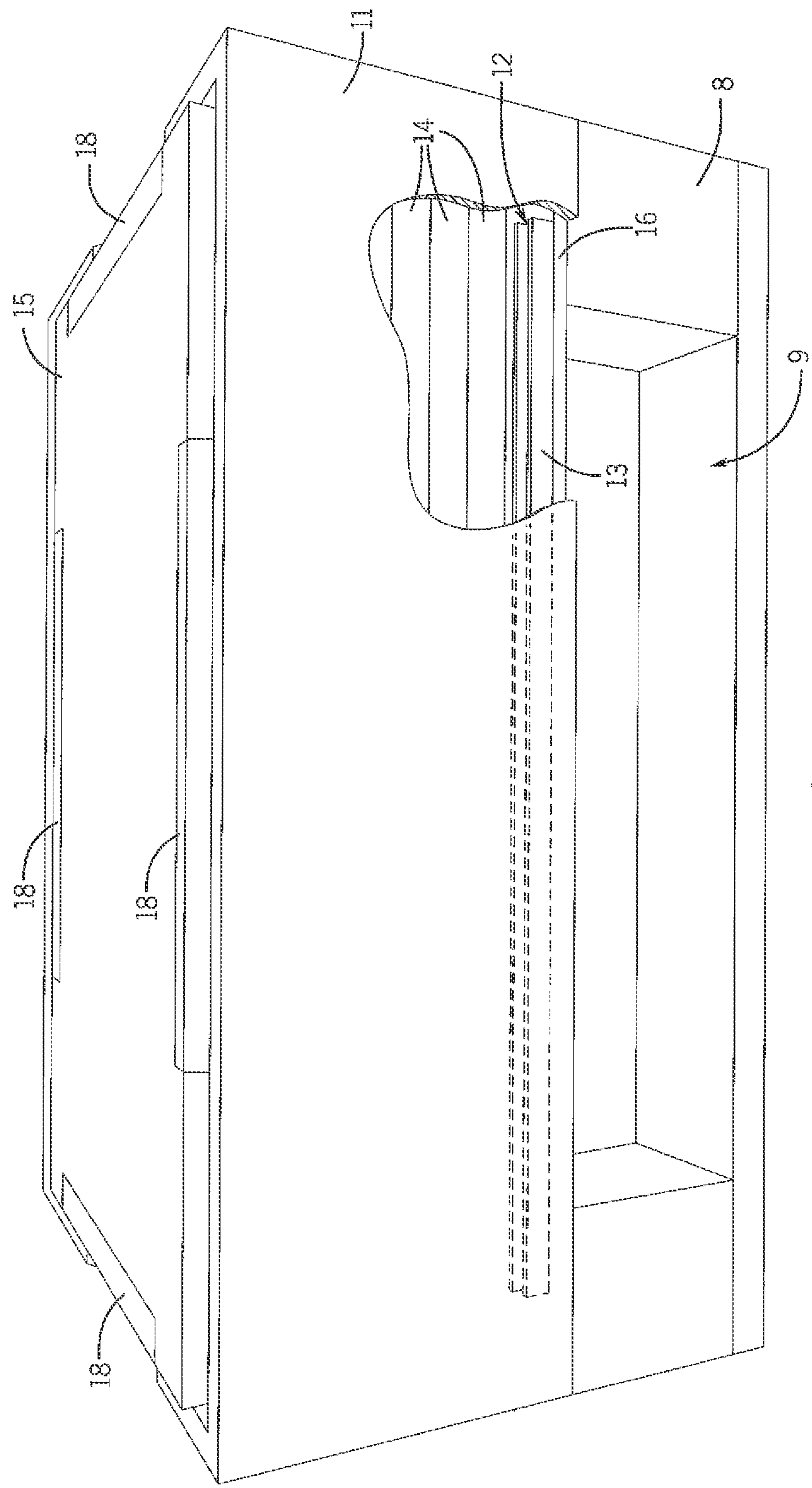



FIG. 2

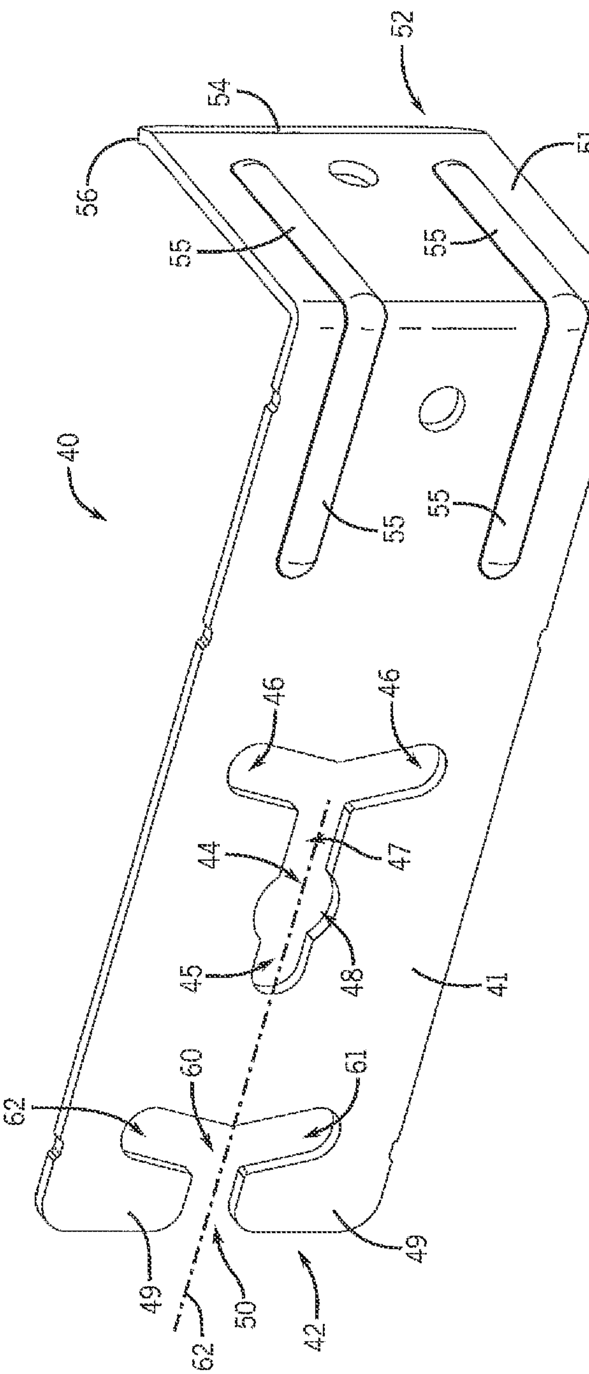

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FIG. 3

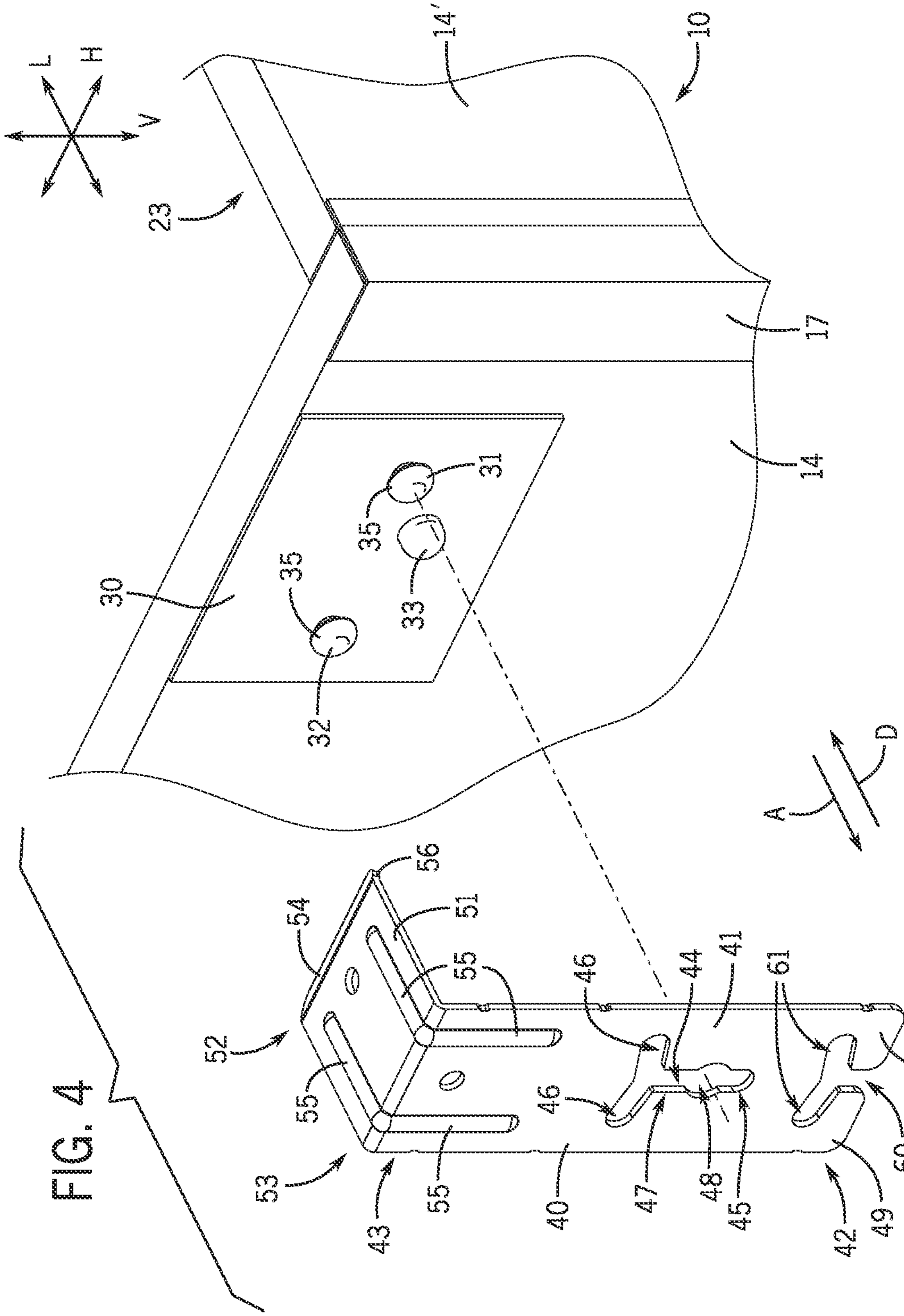

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FIG. 4

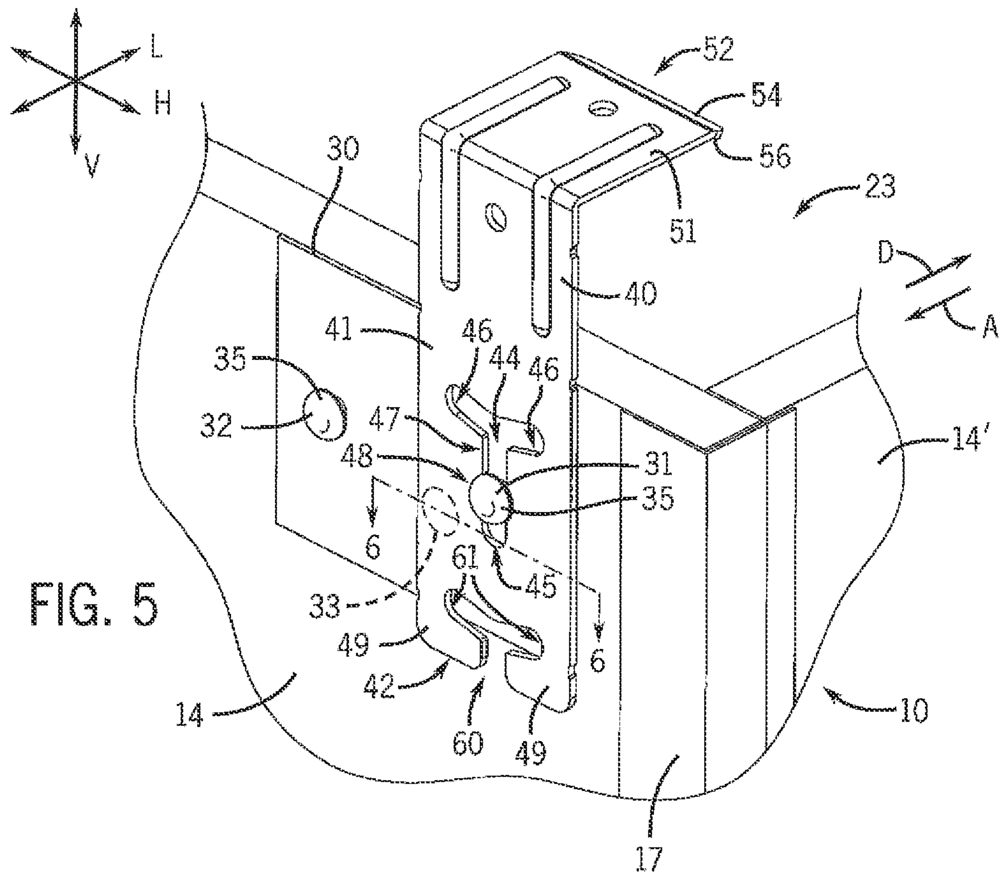
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FIG. 5

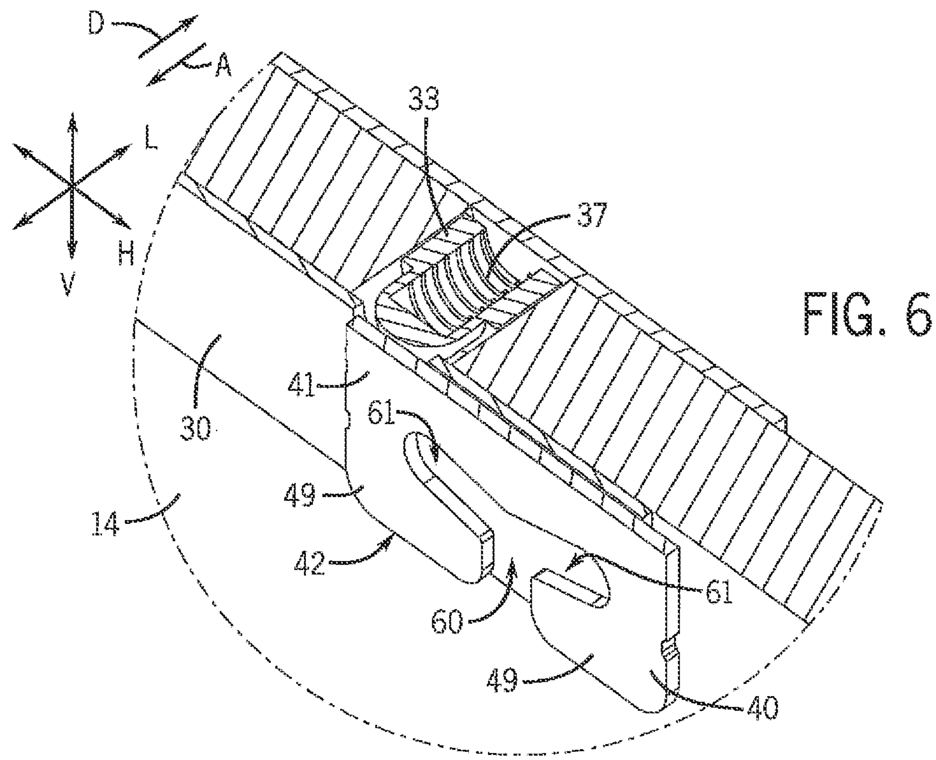
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FIG. 6

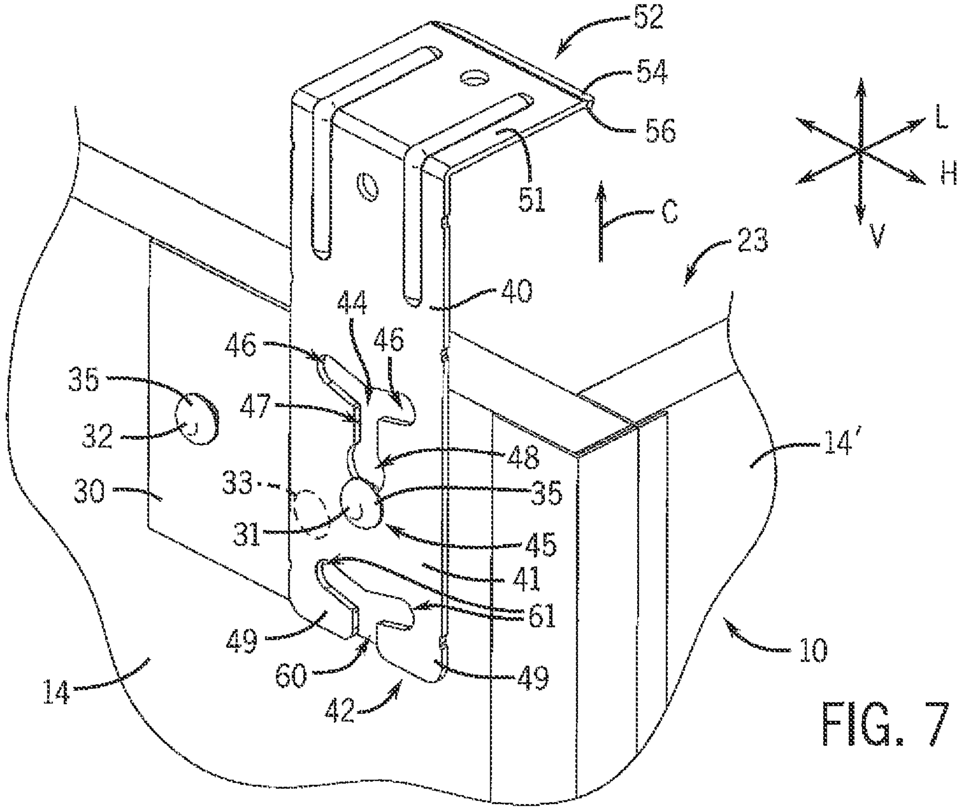
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FIG. 7

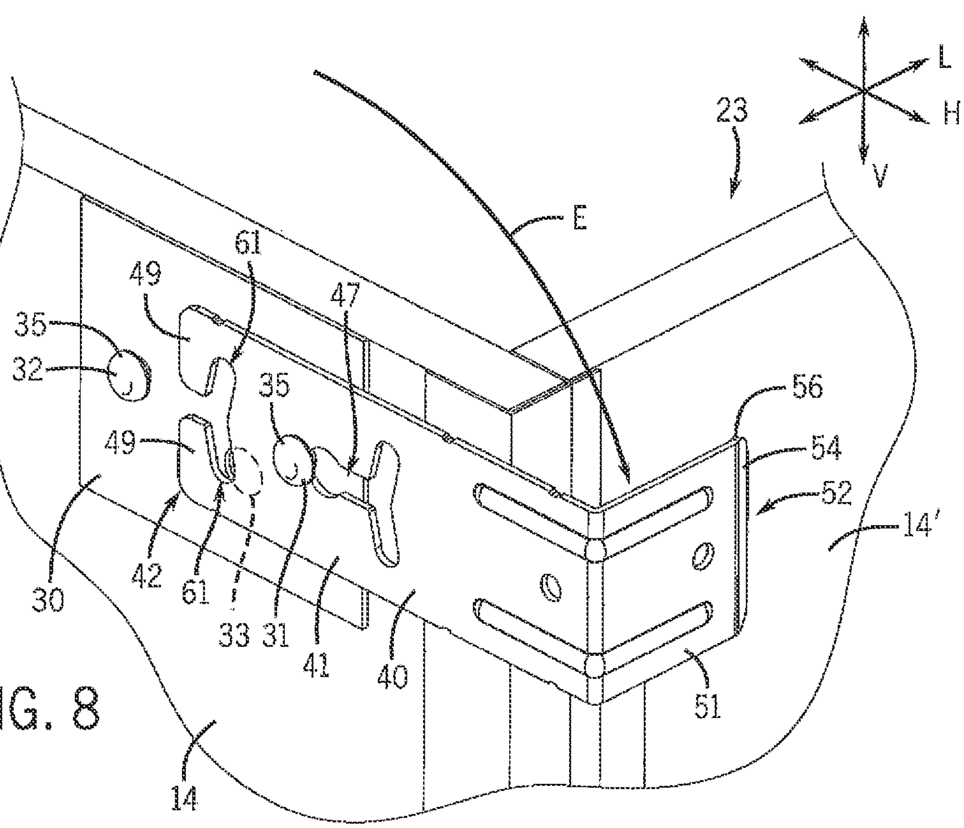
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FIG. 8

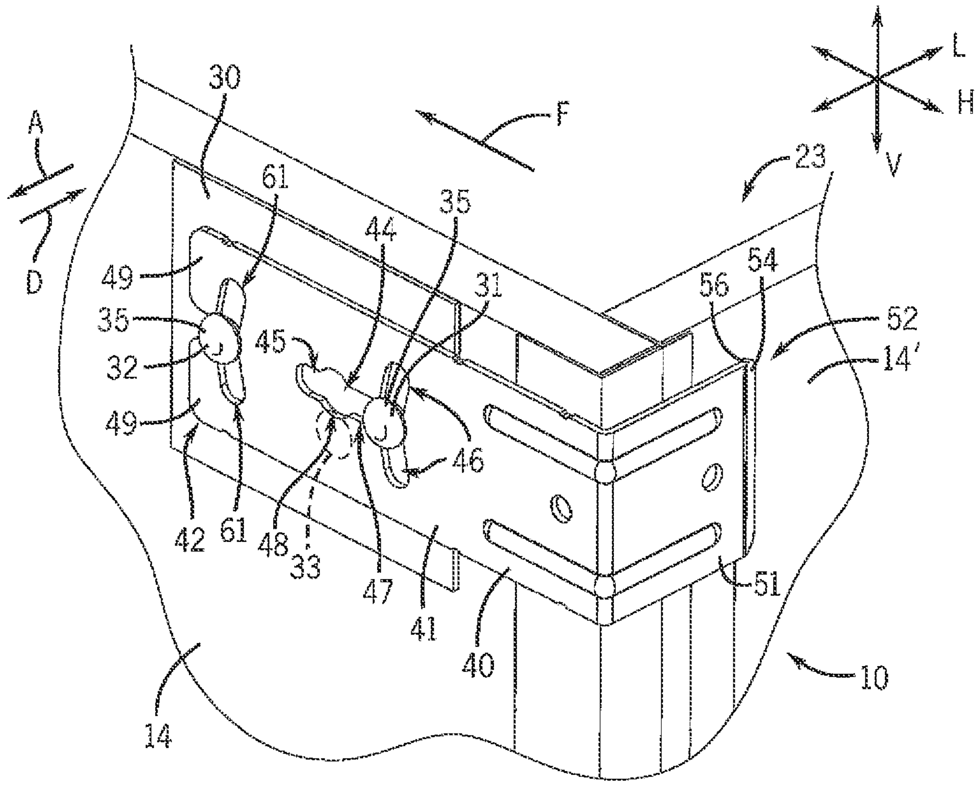
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FIG. 9

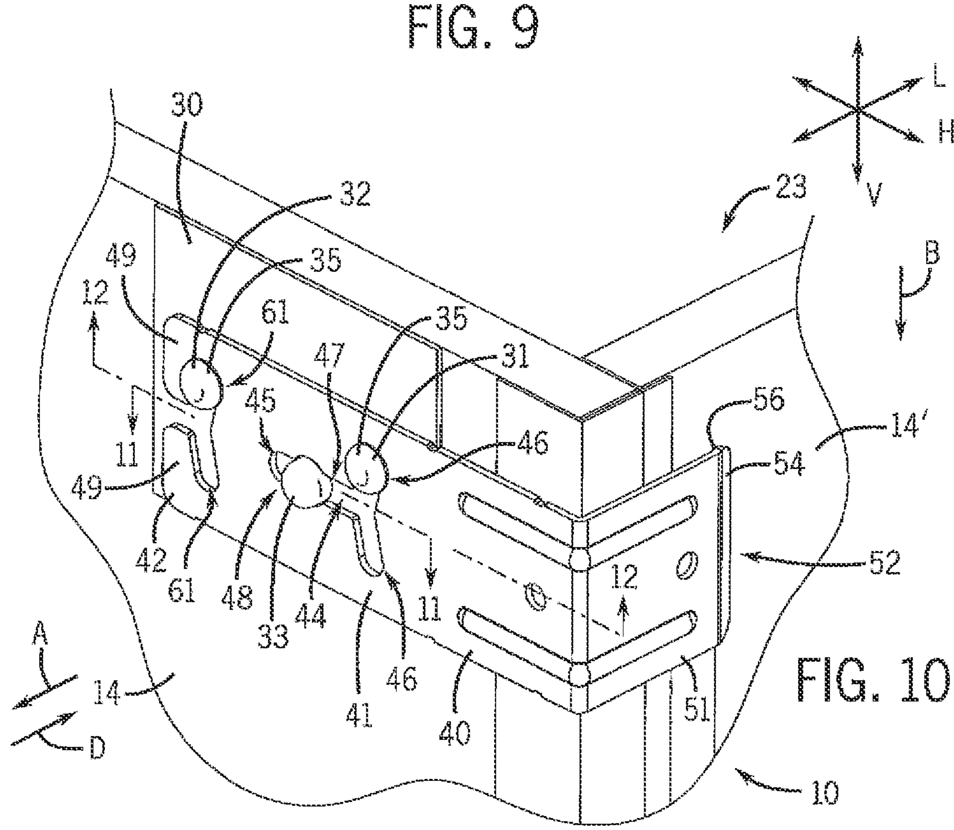
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FIG. 10

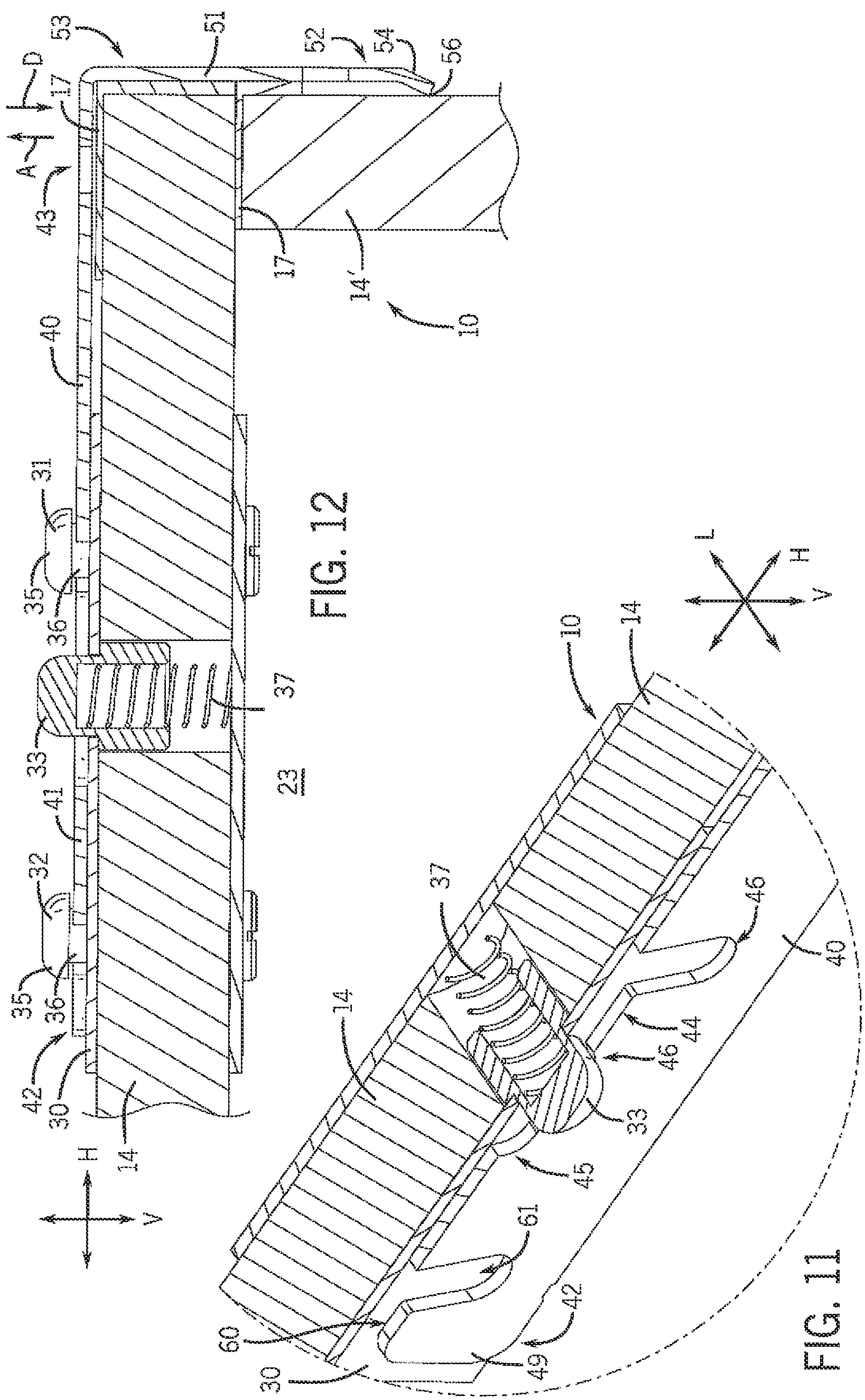

FIG. 12

FIG. 11

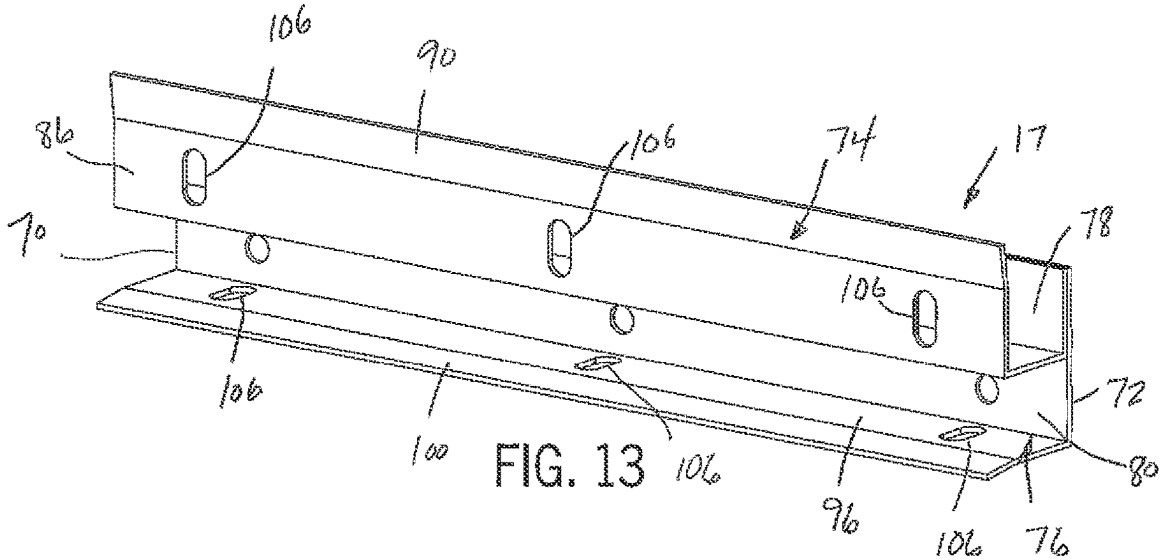


FIG. 13

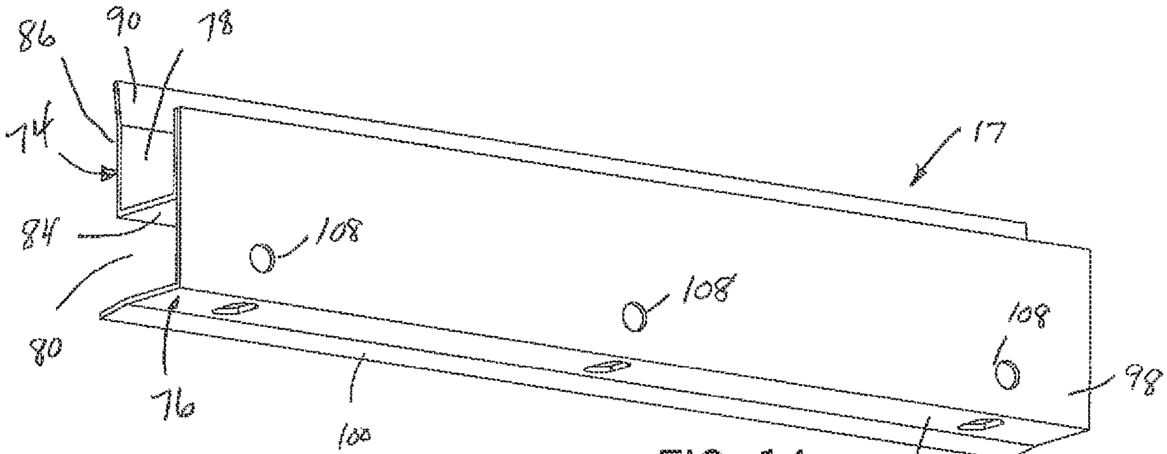


FIG. 14

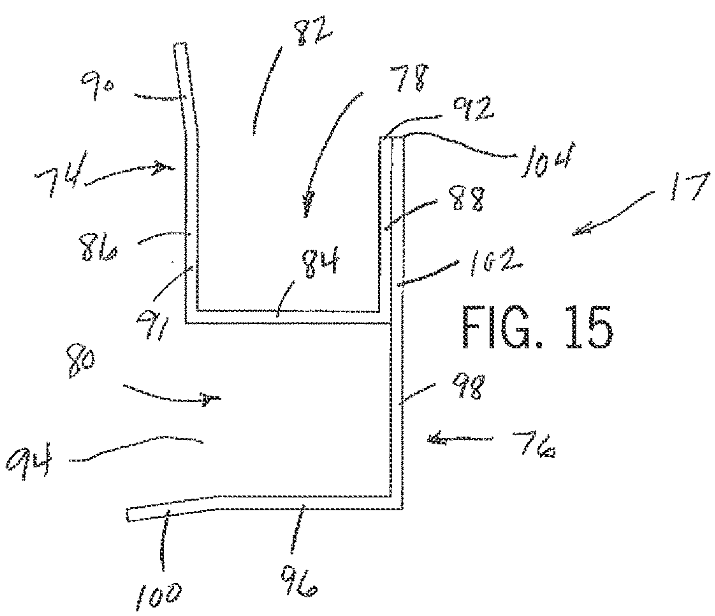


FIG. 15

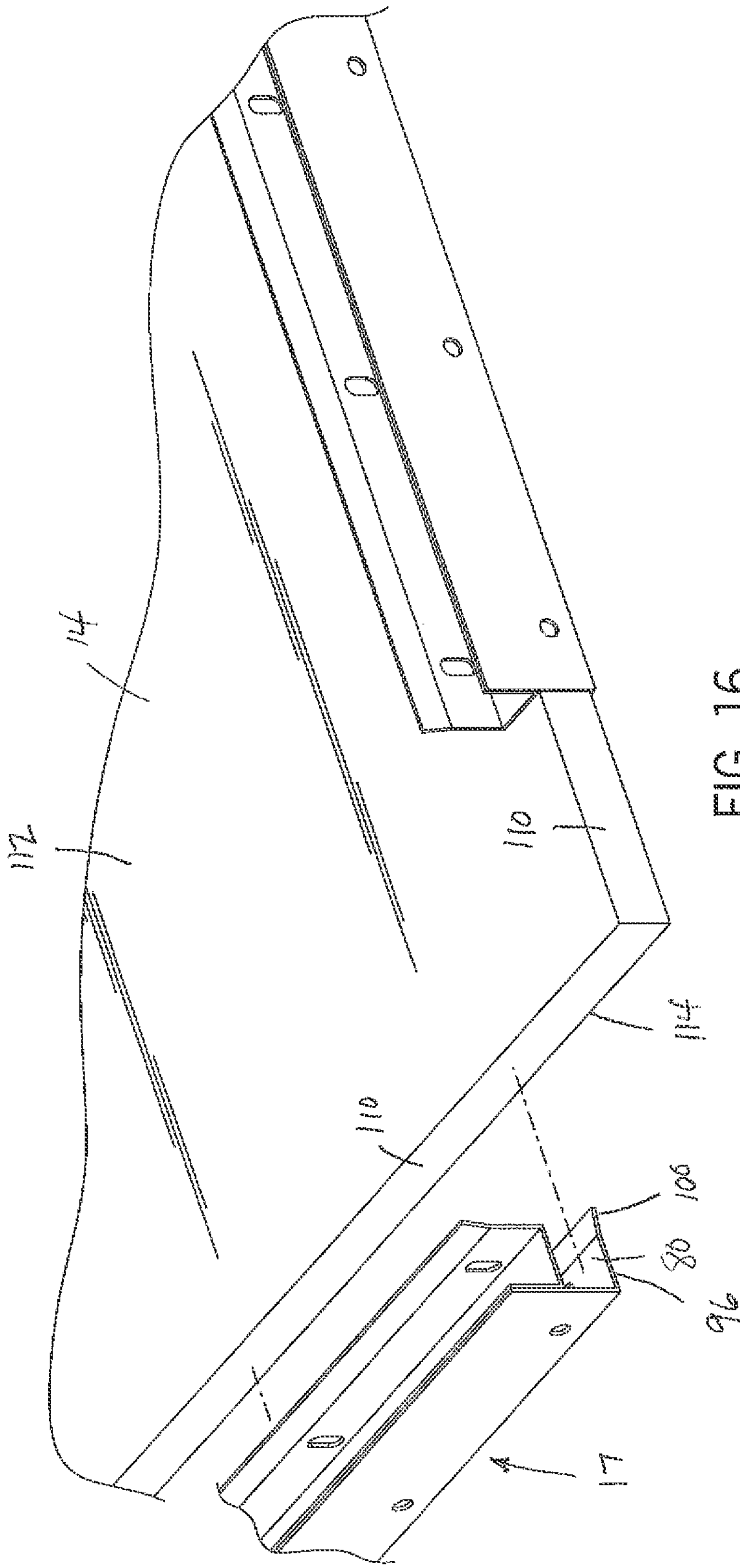


FIG. 16

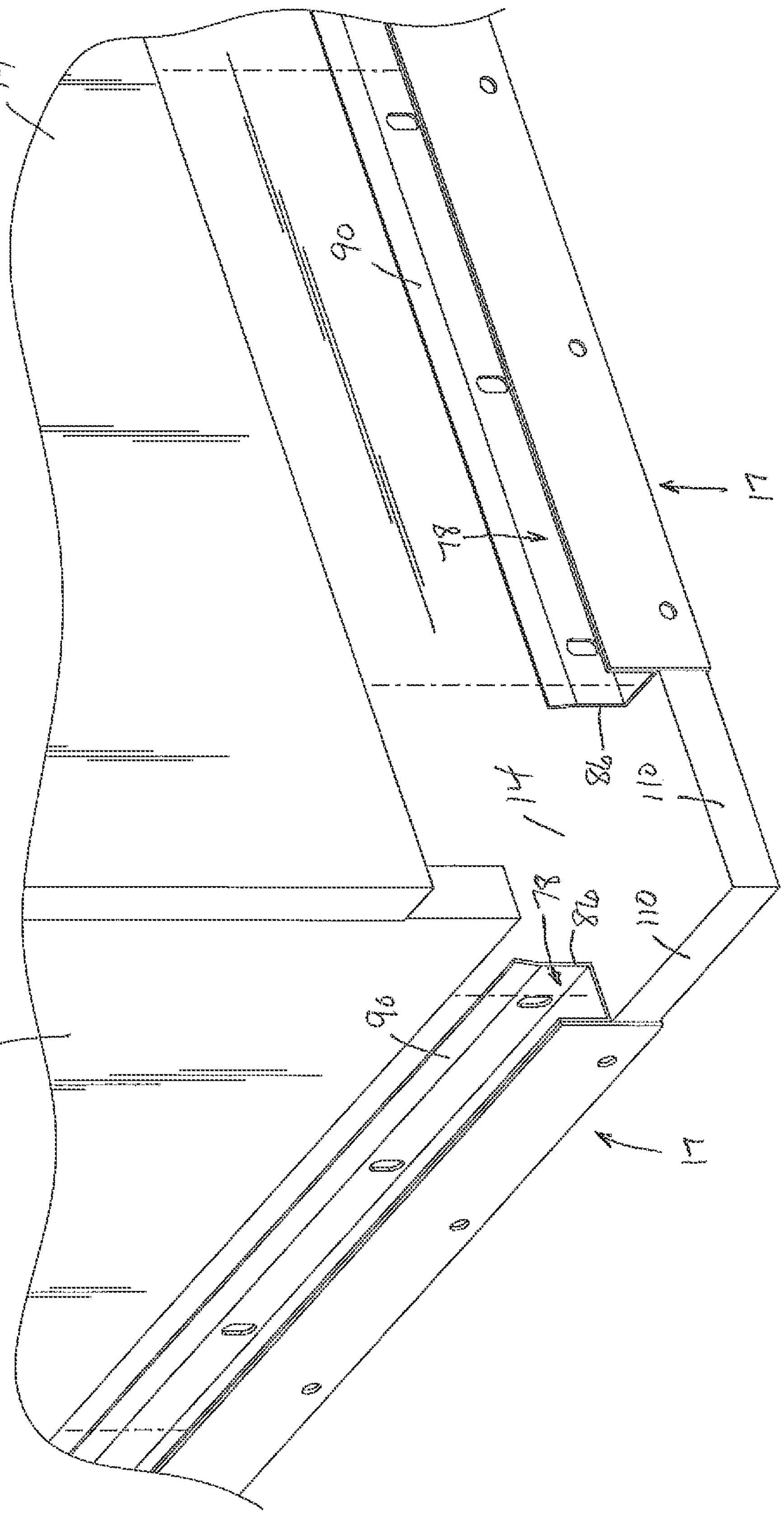


FIG. 17

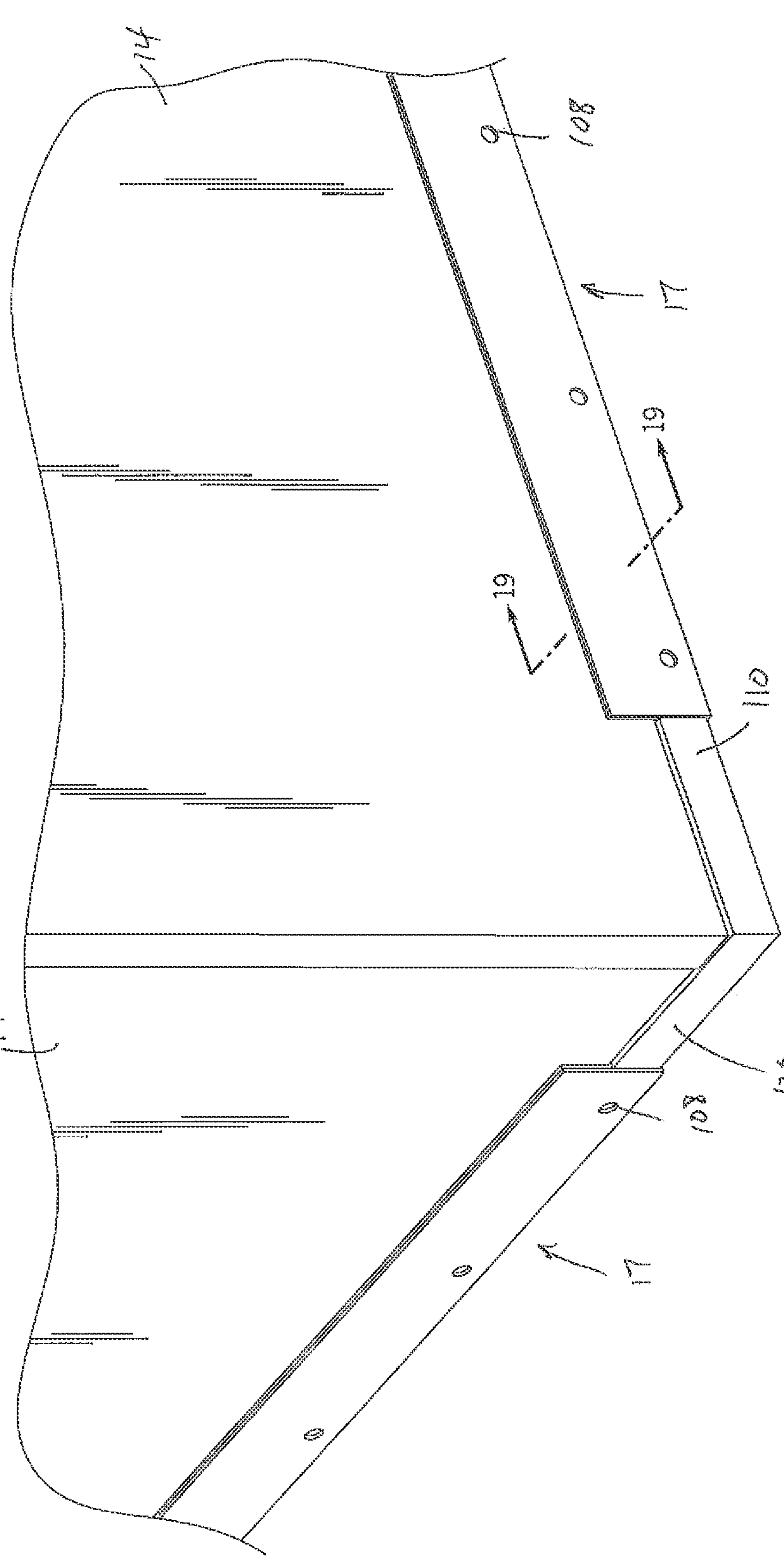


FIG. 18

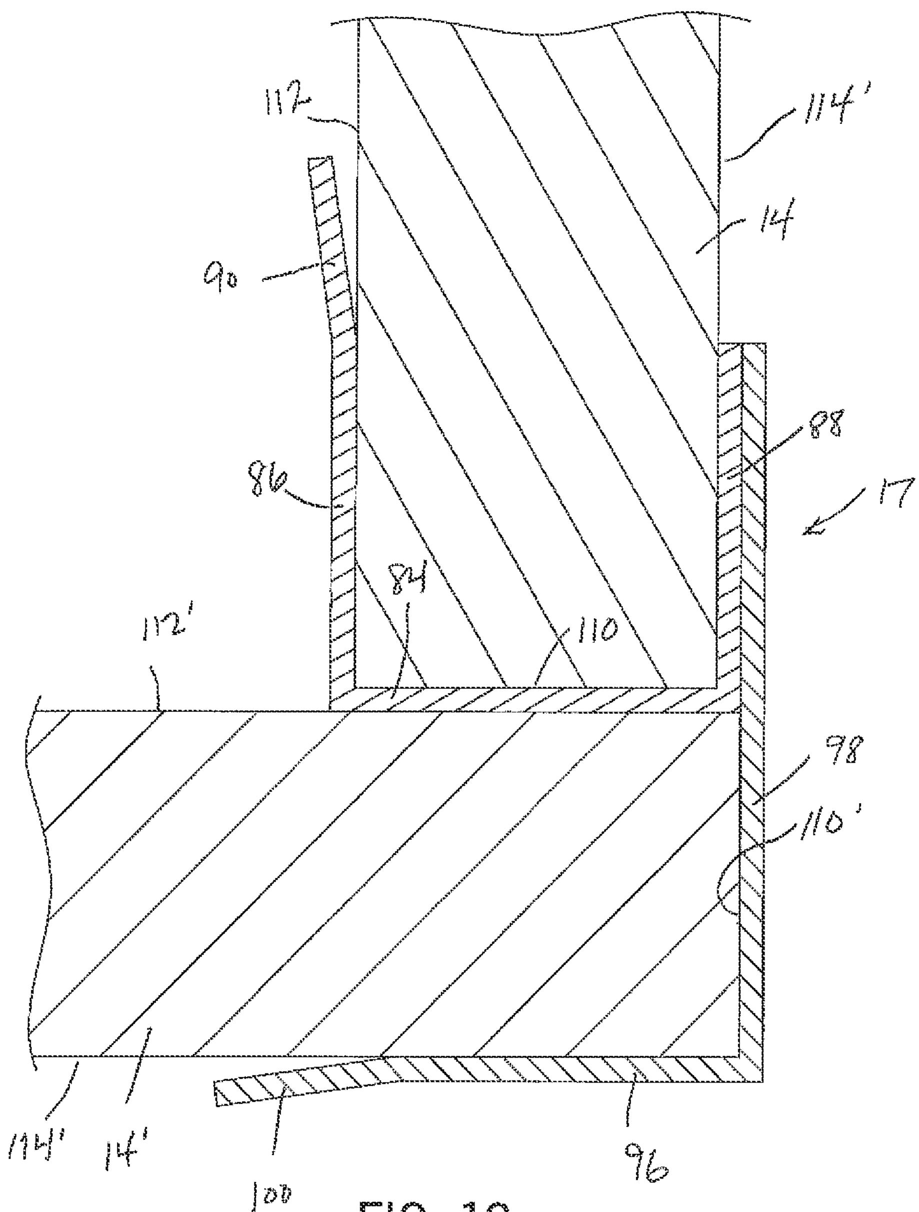


FIG. 19

# COLLAPSIBLE CONTAINERS INCLUDING EDGE ATTACHMENT BRACKETS

## CROSS REFERENCE TO RELATED APPLICATION

The present application is a continuation-in-part and claims priority to U.S. patent application Ser. No. 17/155,683, filed Jan. 22, 2021, the disclosure of which is incorporated herein by reference.

## FIELD

The present disclosure relates to containers used for shipping cargo and specifically to collapsible and reusable containers. More specifically, the present disclosure relates to collapsible containers that includes edge attachment brackets that each join two sides or panels of the collapsible container.

## BACKGROUND

Collapsible containers are commonly used in the shipping industry to transport cargo. Collapsible containers may be reused to reduce waste disposal costs associated with "single-use" shipping containers. Furthermore, collapsible containers may reduce storage space necessary when the containers are not in use.

## SUMMARY

This Summary is provided to introduce a selection of concepts that are further described below in the Detailed Description. This Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

In certain examples, a container includes a first panel with a detent and a pin extending therefrom, a second panel that extends transverse to the first panel, and a bracket removably coupled to the first panel via engagement with the detent and the pin such that the bracket prevents movement of the second panel relative to the first panel. The detent is selectively depressed to thereby decouple the bracket from the first panel.

In certain examples, a container includes a first panel with a detent, a first pin, and a second pin extending therefrom and a second panel that extends transverse to the first panel. A bracket is removably coupled to the first panel via engagement with the detent, the first pin, and the second pin such that a first arm of the bracket is coupled to the first panel and a second arm of the bracket extends along the second panel to thereby prevent movement of the second panel relative to the first panel. The detent is selectively depressed to thereby decouple the bracket from the first panel.

In certain examples, a method for assembling a container includes the steps of positioning a first panel adjacent to the second panel, wherein the first panel has a pin and a detent, positioning a bracket with a slot along the first panel such that the pin is in the slot and the bracket depresses the detent, and moving the bracket relative to the first panel such that the pin is in a leg of the slot and the detent automatically extends into the slot such that the bracket is fixed relative to the first panel.

In one exemplary embodiment of the present disclosure, the container is assembled from a plurality of side panels that each have at least a pair of side edges. The container can include a top panel and a bottom panel that are similar in size and thickness to the side panels. The container includes a plurality of edge brackets that are configured to join the side panels, the top panel and the bottom panel to define an open interior of the container. Each of the edge brackets includes a first channel member and a second channel member that are joined to each other. The first channel member includes a first receiving channel while the second channel member includes a second receiving channel. The first and second channels are each sized to receive the side edge of one of the side panels. The first and second channels are oriented perpendicular to each other such that when the side panels are received by the edge bracket, the side panels are perpendicular to each other.

Various other features, objects, and advantages will be made apparent from the following description taken together with the drawings.

## BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure is described with reference to the following Figures. The same numbers are used throughout the Figures to reference like features and like components.

FIG. **1** is a perspective view of an example container with side panels assembled and received on a bottom panel and edge brackets securing the side panels to each other;

FIG. **2** is a perspective view of an example container system with panels stacked in the base;

FIG. **3** is a perspective view of an example bracket according to the present disclosure;

FIG. **4** is an exploded view showing the bracket removed from a container panel;

FIG. **5** is a partial view showing the attachment of the bracket to one of the container panels;

FIG. **6** is a cross-sectional view of the bracket and the panel taken along line **6-6** on FIG. **5**;

FIG. **7** is a view of the bracket in an upward position;

FIG. **8** is a view of the bracket in a rotated position;

FIG. **9** is a view of the bracket in a retracted position;

FIG. **10** is a view of the bracket in the locked position;

FIG. **11** is a section view of the bracket and the panel taken along line **11-11** of FIG. **10**;

FIG. **12** is a section view of the bracket and panel taken along line **12-12** of FIG. **10**;

FIG. **13** is a front perspective view of an edge bracket in accordance with an exemplary embodiment of the present disclosure.

FIG. **14** is a back perspective view of the edge bracket of the exemplary embodiment of the present disclosure;

FIG. **15** is a side view of the edge bracket;

FIG. **16** is an exploded view showing the edge bracket and one of the plurality of panels;

FIG. **17** is an exploded view showing two panels being received by a pair of edge brackets mounted to another of the panels;

FIG. **18** is a view showing the two panels mounted to a base panel using the pair of edge brackets; and

FIG. **19** is a section view taken along line **19-19** of FIG. **18**.

## DETAILED DESCRIPTION

FIGS. **1-2** depict an example container **10** of the present disclosure in which cargo is received. As depicted in FIG. **1**, the container **10** is formed by a plurality of panels **14**, **15**, **16** that are stacked within a base **11**. When assembled, the container **10** defines an internal cavity **23** (FIG. **4**) in which cargo is contained. Note that the container **10** generally extends along a vertical axis (see arrow V) between a top and a bottom, a lateral axis (see arrow H) between a front and a back, and a longitudinal axis (see arrow L) between a first side and an opposite second side. The containers **10** are formed by any suitable material such as metal, plywood, corrugated cardboard, and solid wood.

Also note that although FIG. **1** depicts only one side panel **14** and a top panel **15**, the container **10** includes additional side panels **14** (see FIG. **4**) and a bottom panel **16** (see FIG. **2**). The shape of the container **10** can vary, and in one example, the container **10** has a total of four side panels **14** such that the container **10** has a generally a rectangular prism.

The side panels **14** are joined to each other to defined the container by a series of edge brackets **17** that both protect the ends and edges of the side panels **14** and provide the means for attaching the side panels **14** to each other and to the top and bottom panels **15**, **16**. The edge brackets **17** are coupled to the side panels **14** and the top and bottom brackets **15**, **16** with adhesives or fasteners, such as screws. The edge brackets **17** are L-shaped, U-shaped or other similar configuration and are formed of any suitable material such as metal and plastic. Further details of the edge brackets **17** are shown in FIG. **13-19** as will be fully described below.

Each side panel **14** can also include one or more mounting plates **30** that each include a first pin **31**, a second pin **32**, and a detent **33** extending therefrom. The mounting plates **30** are on an exterior surface of the side panel **14** that is opposite an interior surface that faces an internal cavity **23** (FIG. **4**) formed by container **10**. As such, the first pin **31**, the second pin **32**, and the detent **33** extend outwardly in a forward or first direction (see arrow A) away from the exterior surface of the container **10**. Each pin **31**, **32** has a pinhead **35** and a shaft **36** (FIG. **12**) that extends between the mounting plate **30** and the pinhead **35**. The pinhead **35** is enlarged relative to the cross-section or diameter of the shaft **36**. The pins **31**, **32** are aligned with each other in the lateral direction and the detent **33** is offset from the pins **31**, **32** (e.g., the pins **31**, **32** are positioned along a common line and the detent **33** is not positioned along the common line). For instance, the detent **33** is vertically offset from pins **31**, **32**. In certain examples, the pins **31**, **32** are fixed on the side panel **14**. In certain examples, the number of pins can vary (e.g., only one pin extends from the mounting plate, four pins extend from the mounting plate).

The number of mounting plates **30** on each side panel **14** can vary, and in one example, two mounting plates **30** are coupled to the top portions of two opposing side panels **14**. Note that in certain examples, pins **31**, **32** and the detent **33** are coupled directly to the exterior surface of the side panel **14** such that the mounting plate **30** can be excluded. Brackets **40** (described further herein) are coupled to each mounting plate **30** to thereby prevent movement of adjacent side panels **14** (described further herein) and/or secure the shape of the container **10**. In one example, the brackets **40** prevent movement of the side panels **14** in an outward direction (see arrows J on FIG. **1**) relative to the center of the internal cavity **23** (FIG. **4**).

The top panel **15** rests on top of the side panels **14** and defines the top of the container **10**. The top panel **15** has one or more lips **18** that extend from the sides or edges of the top panel **15**. Thus, when the top panel **15** is placed on top of the side panels **14**, the lips **18** extend in a second or downward direction (see arrow B on FIG. **1**) and along the exterior surfaces of the side panels **14**. Therefore, the lips **18** prevent the top portions of the side panels **14** from bowing outwardly. The lips **18** are coupled to the top panels **15** with adhesives or fasteners such as screws. The lips **18** are L-shaped and formed with any suitable material such as metal and plastic.

The bottom panel **16** (FIG. **2**) rests on the base **11** and has a plurality of channels **12** that receive bottom ends of the side panels **14**. Accordingly, the side panels **14** are prevented from inwardly moving or bowing. That is, when the side panels **14** are in the channels **12**, the side panels do not move in an inward direction (see arrow I on FIG. **1**) relative to the center of the internal cavity **23** (FIG. **4**). The bottom panel **16** defines the bottom of the container **10** and the side panels **14** are vertically supported when the side panels **14** are in the channels **12**. The channels **12** are defined U-shaped support members **13** that vertically upwardly extend toward the top of the container **10**. In other examples, the channels **12** are recessed in the bottom panel **16**.

The bottom panel **16** rests on the base **11**, and thus, the container **10** rests on the base **11** when constructed. The base **11** has a recessed chamber **20** in which the container **10** sits such that the bottom panel **16** and the lower portions of the side panels **14** are in the chamber **20**. Sidewalls **21** of the base **11** define the sides of the chamber **20**. Furthermore, as depicted in FIG. **2**, when the container **10** is disassembled (e.g., the panels **14**, **15**, **16** are decoupled from each other and no longer form the container), the chamber **20** advantageously receives the panels **14**, **15**, **16**. That is, when the container **10** is disassembled or collapsed, the panels **14**, **15**, **16** are advantageously positioned in the chamber **20** in a stacked configuration (see FIG. **2**). Accordingly, the panels **14**, **15**, **16** can be easily stored in the base **11**. Optionally, the base **11** rests on a pedestal **8** having fork-receiving openings **9** or a convention pallet (not shown). Thus, the base **11** and the container **10** (either assembled or disassembled) is easily transported on the pedestal **8**. Note that in certain examples, the base panel **16** is excluded and the channels **12** are in the base **11**.

As noted above, a bracket **40** is selectively coupled to each mounting plate **30** on the side panels **14**, to thereby prevent movement of adjacent side panels **14** (described further herein). Referring now to FIG. **3**, the bracket **40** is depicted in greater detail. The bracket **40** has a first arm **41** and a second arm **51** that is fixedly coupled to the first arm **41** such that the first arm **41** extends transverse to the second arm **51**. The first arm **41** has a free first end **42** and an opposite second end **43** that is coupled to a second end **53** of the second arm **51**. The second arm **51** also has a free first end **52** with an end portion **54** that extends transverse to the remaining portion of the second arm **51**. Thus, the end portion **54** has an edge **56** that is configured to engage the exterior surface of a side panel **14** when the bracket **40** is coupled to the mounting plate **30**. The second arm **51** includes stiffening elements **55** (e.g., welds, recessed channels such that the material on second arm **51** opposite the channel "bulges" outwardly) that stiffen the second arm **51** and prevent bending of the second arm **51**. The first arm **41** also includes stiffening elements **55** that stiffen the first arm **41** and prevent bending of the first arm **41**. Certain stiffening elements **55** of the first arm **41** and the second arm **51** are contiguous with each other such that these stiffening elements **55** prevent inadvertent bending of the first arm **41** relative to the second arm **51**.

The first arm **41** has a first slot **44** and a second slot **60** that is spaced apart from the first slot **44**. The first slot **44** has a head **45**, a pair of opposing legs **46**, and a body **47** that extends between the head **45** and the legs **46**. The body **47** has an enlarged section **48** through which the pinhead **35** of the first pin **31** passes. That is, the width or diameter of the enlarged section **48** is greater than or equal to the width or diameter of the pinhead **35** of the first pin **31**. The width of the other sections of the body **47**, the head **45**, and the legs **46** is less than the width or diameter of the pinhead **35** of the first pin **31**. Thus, the pinhead **35** of the first pin **31** does not pass through the other sections of the body **47**, the head **45**, and the legs **46** and the pinhead **35** only passes through the enlarged section **48** (described further herein). Each leg **46** extends transverse to the body **47** and an angle is formed between the body **47** and each leg **46**.

The second slot **60** is at the first end **42** of the first arm **41** such that the first end **42** of the first arm **41** has an opening **50** and two opposing prongs **49**. Thus, the second pin **32** can pass between the prongs **49** into the second slot **60**. The second slot **60** has a pair of opposing legs **61**. Note that the width of the second slot **60** is less than the width or diameter of the pinhead **35** of the second pin **32**. Thus, the second pin **31** only passes into or out of the second slot **60** when the bracket **40** is moved such that the second pin **31** is moved between the prongs **49**. Each leg **61** extends away from each other and transverse to an axis **62**. An angle is defined between the axis **62** and each leg **61**. In certain examples, the angle defined between the axis **62** and one of the legs **61** is equal to the angle defined between the body **47** and one of the legs **46**. In certain examples, the one leg **46** of the first slot **44** extending in the same direction as one leg **61** of the second slot **60**. Note that the first arm **41** extends along the axis **62**

FIGS. **4-15** depict an example sequence for coupling the bracket **40** to the mounting plate **30** that is on one of the side panels **14**. Accordingly, as noted above, the bracket **40** prevents an adjacent side panel **14'** from moving relative to the side panel **14** on which the mounting plate **30** is coupled. A person of ordinary skill in the art will recognize that the steps noted below may be combined with other steps. In certain examples, the side panels **14**, **14'** abut each other.

FIG. **4** depicts the one side panel **14** adjacent to another side panel **14'**. The bracket **40** is spaced apart from the mounting plate **30** such that the enlarged section **48** of the first slot **44** is aligned with the first pin **31**. The operator moves the bracket **40** in a fourth direction (see arrow D) toward the mounting plate **30** such that the first pin **31** passes through the enlarged section **48** of the first slot **44**.

Referring to FIGS. **5-6**, as the first pin **31** passes through the enlarged section **48** of the first slot **44**, the bracket **40** acts on and moves the detent **33** in the fourth direction (see arrow D). That is, the bracket **40** forces the detent **33** in the fourth direction (see arrow D) against a spring force exerted by a spring **37** (FIG. **6**) that normally biases the detent **33** in a first direction (see arrow A). Thus, the spring **37** is compressed as the detent **33** is moved in the fourth direction (see arrow D). Note that the detent **33** is depicted in dashed lines on FIG. **5**.

Referring to FIG. **7**, after the first pin **31** passes through the enlarged section **48** of the first slot **44**, the operator upwardly slides the bracket **40** in the third direction (see arrow C) such that the bracket **40** slides along the first pin **31** until the first pin **31** is at the head **45**. Note that the pinhead **35** of the first pin **31** slides along the exterior surface of the bracket **40** while the shaft **36** (FIG. **12**) of the first pin **31** slides in the first slot **44**. Thus, the pinhead **35** prevents the bracket **40** from moving the first direction (see arrow A) away from the side panel **14**. Note that the bracket **40** continues to act on the detent **33** as described above while the operator upwardly slides the bracket **40** in the third direction (see arrow C).

FIG. **8** depicts the operator pivoting the bracket **40** about the first pin **31** in a first pivot direction (see arrow E; e.g., clockwise direction) such that the second arm **51** of the bracket **40** is along the adjacent side panel **14'**. Note that the bracket **40** continues to act on the detent **33** as described above while the operator pivots the bracket **40**.

The operator then slides the bracket **40** in a fifth direction (see arrow F), as depicted in FIG. **9**, until the second pin **32** is in the second slot **60**, the first pin **31** is near the legs **46** of the first slot **44**, and the second arm **51** of the bracket **40** is along the adjacent side panel **14'**. In certain examples, the edge **56** of the second arm **51** contacts the exterior surface of the adjacent side panel **14'**. Note that the pinhead **35** of the second pin **32** is along the surface of the bracket **40** while the shaft **36** (FIG. **12**) of the second pin **32** is in the second slot **60**. Thus, the pinhead **35** of the second pin **32** also prevents the bracket **40** from moving in the first direction (see arrow A) away from the side panel **14** (e.g., the pinheads **35** of the first pin **31** and the second pin **32** retain the bracket **40** on the panel **14**). Note that the bracket **40** continues to act on the detent **33** as described above while the operator slides the bracket **40** in the fifth direction (see arrow F).

Referring now to FIGS. **10-12**, the operator completes the coupling sequence (e.g., to fully secure or lock the bracket **40** into position on the mounting bracket **40**) by sliding the bracket **40** generally downwardly (e.g., see second direction indicated by arrow B) such that the first pin **31** is in one of the legs **46** and the second pin **32** is in one of the legs **61**. As the operator slides the bracket **40**, the detent **33** is moved by the spring **37** (FIG. **12**) in the first direction (see arrow A) such that the detent **33** extends through the enlarged section **48** of the first slot **44**. As noted above, the detent **33** is offset from the pins **31**, **32** (e.g., the detent **33** is vertically offset from the pins **31**, **32**) and accordingly, the bracket **40** is fixed in place on the mounting bracket **40**. Thus, the second arm **51** of the bracket **40** is also fixed relative to the adjacent side panel **14'** and thus, the bracket **40** creates a braced connection between the side panel **14** and the adjacent side panel **14'** such that the bracket prevents the side panels **14**, **14'** from moving relative to each other.

The operator decouples the bracket **40** from the mounting bracket **40** (such as when the container **10** is to be disassembled or collapsed) by pressing the detent **33** in the fourth direction (see arrow D on FIG. **10**) against the spring force applied by the spring **37** to the detent **33**. Accordingly, the operator can then move (e.g., slide, pivot) the bracket **40** in different directions to decouple the bracket **40** from the mounting plate **30**. Note that a person of ordinary skill in the art will recognize that the directions the operator must moves the bracket **40** to decouple to the bracket **40** from the mounting plate **30** are generally opposite the directions noted above with respect to the example coupling sequence for coupling the bracket **40** to the mounting plate **30**.

Note that the legs **46**, **61** of the slots **44**, **60**, respectively, are advantageously arranged such that the bracket **40** can be used on any side panel **14** of the container. For instance, FIGS. **4-12** depict the bracket **40** on a right side of the container **10**. However, the bracket **40** could be "flipped" and used on the opposite, left side of the container **10**.

As illustrated in FIGS. **1**, **4** and **12**, adjacent side panels **14** of the container **10** are joined to each other by an edge bracket **17** that is positioned at the perpendicular intersection between the side panels **14**. The edge bracket **17** can also be used to join the side panels **14** to the top panel **15**, as will be described below. The edge brackets **17** are each designed to allow the side panels **14** to be assembled to create the side walls of the container **10** and to define an open interior for the container. FIGS. **13-15** illustrate the details of the edge brackets **17** that are used to join the four side panels **14** used to form the collapsible container **10** of the present disclosure.

The edge bracket **17** shown in FIGS. **13-15** is formed from a metallic material and extends along a longitudinal length between a first end **70** and a second end **72**. However, the edge bracket **17** could be formed from other durable materials, such as plastic or nylon. When the edge bracket **17** is used to form the corners of the container between adjacent side panels, the longitudinal length of the edge bracket **17** generally corresponds to the height of each of the side panels **14**. In such embodiment, the edge bracket **17** extends along the entire height of each of the side panels when the container is constructed as shown in FIG. **1**.

The edge bracket **17** is formed from two separate components that are joined to each other to create the edge bracket **17**. In the embodiment shown, a first channel member **74** and a second channel member **76** are joined to each other to form the integrated edge bracket **17**. The first channel member **74** is shaped and configured to define a first receiving channel **78** while the second channel member **76** is shaped and configured to define a second receiving channel **80**. The first and second receiving channels **78**, **80** each extend along the entire longitudinal length of the edge bracket **17** between the first end **70** and the second end **72**. As can be seen in FIG. **15**, the first receiving channel **78** and the second receiving channel **80** are oriented such that the first and second receiving channels **78**, **80** are perpendicular to each other.

The first receiving channel **78** includes an open end **82** and is defined by a back wall **84**, a first side wall **86** and a second side wall **88**. The first and second side walls **86**, **88** extend from opposite edges of the back all **84** and extend parallel to each other and perpendicular to the back wall **84**. The width of the first receiving channel **78** is thus defined by the distance between the first side wall **86** and the second side wall **88**. As shown in FIG. **15**, the first side wall **86** includes an extended portion **90** that is angled slightly away from the lower portion **91** of the first side wall **86** to expand the effective width of the open end **82** above the top edge **92** of the second side wall **88**. The angled orientation of the extended portion **90** allows for a side edge of one of the side panels to enter into the first receiving channel **78** more easily as will be described in greater detail below.

The second channel member **76** partially defines the second receiving channel **80**, which includes an open end **94**. The second receiving channel **80** is defined by a side wall **96**, a back wall **98** and the back wall **84** of the first channel member **74**. The width of the second receiving channel **80** is defined by the distance between the side wall **96** and the back wall **84**. The width of the second receiving channel **80** corresponds to the width of the first receiving channel **78**, which in turn corresponds to the thickness of the side panels the form the container of the present disclosure. The side wall **96** includes an extended portion **100** that angles outward away from the open end **94** to expand the width of the open end **94** to allow the side edge of one of the side panels to more easily be received within the second receiving channel **80**.

As most clearly shown in FIG. **15**, the back wall **98** of the second receiving channel **76** extends past the back wall **84** of the first channel member such that the extended portion **102** of the back wall **98** is adjacent to the second side wall **88**. The top edge **104** of the extended portion **102** is generally aligned with the top edge **92** of the second side wall **88**. As can be seen in FIG. **15**, the second side wall **88** is securely joined to the extended portion **102** of the back wall **98** such that the first channel member **74** is joined to the second channel member **78**. In an embodiment in which the first channel member **74** and the second channel member **76** are each formed from a metal material, the first and second channel members **74**, **76** can be joined to each other using any type of metal joining process, such as using an adhesive or a weld. In an embodiment in which the first channel member **74** and the second channel member **76** are formed from a plastic or nylon material, plastic and nylon joining methods, such as an adhesive or a heat weld, can be used to join the first and second channel members **74**, **76**. In each embodiment, the first and second channel members **74**, **76** are securely joined to each other to create the edge bracket **17**.

Referring to FIGS. **13** and **14**, the first side wall **86** and the side wall **96** are each formed having a plurality of attachment openings **106** evenly spaced between the first end **70** and the second end **72** of the edge bracket **17**. The attachment opening **106** are each designed to receive a connector, such as a screw or nail, to securely join the edge bracket **17** to the pair of joined side panels in a manner as will be described below. In addition to the attachment openings **106**, the edge bracket **17** can include another series of attachment openings **108** formed in the lower portion of the back wall **98**. The attachment openings **108** provide access to the second receiving channel **80** and the side edge of the side panel received therein. The series of attachment openings **106** and **108** allow the edge bracket **17** to be securely attached to a pair of side panels when the side panels are oriented perpendicular to each other to form the container.

FIG. **16** illustrates one of the side panels **14** that is used to form the collapsible container. The side panel **14** includes four side edges **110**, two of which are shown in FIG. **16**. The side edges **110** each have a thickness defined between the upper face **112** and the lower face **114**. The thickness of the side edge **110** corresponds to the width of the second receiving channel **80** of the edge bracket **17**. As illustrated in FIG. **16**, the edge bracket **17** can be pressed onto the side edge **110** such that the edge bracket **17** moves from the detached position to a mounted position relative to the side panel **14**, as shown in FIG. **16**. As described previously, the side wall **96** includes an outer portion **100** that is angled slightly to expand the open end to facilitate the insertion of the side edge **110** into the second receiving channel **80**.

After the pair of edge brackets **17** are received on the side edges **110** of the side panel **14**, two more side panels **14** can be inserted into the respective first receiving channels **78**. Each of the first receiving channels **78** includes the open outer end which is expanded by the angled upper portion **90** of the first side wall **86**. Once the side panels are received as shown in FIG. **18**, connectors can be used in the attachment openings **108** to secure the edge bracket **17** to the side panel **14**.

FIG. **19** is a section view that illustrates the use of the edge brackets **17** to securely hold the pair of side panels **14** and **14'** in a perpendicular orientation to each other. In this position, the side edge **110** of the first side panel contacts the back wall **84** and the pair of spaced side walls **86** and **88**. The upper portion **90** is positioned at an angle outward such that it is spaced from the face **112** of the side panel. The second side panel **14'** includes a side edge **110'** that contacts the back wall **98**. The face **112'** is in contact with the back wall **84** while the face **114'** is in contact with the side wall **96**. The outer portion **100** extends away from the face **114'** to facilitate the insertion of the side panel **14'**.

When the pair of side panels **14** are inserted and installed as shown in FIG. **19**, connectors can then be used to join the first and second channel members **74**, **76** to the pair of side panels. The connectors are positioned to extend through the attachment openings **106**, which are best shown in FIG. **13**. In this manner, the edge brackets **17** can be each used to join a pair of perpendicular side panels.

In the present description, certain terms have been used for brevity, clarity, and understanding. No unnecessary limitations are to be inferred therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed. The different apparatuses, systems, and method steps described herein may be used alone or in combination with other apparatuses, systems, and methods. It is to be expected that various equivalents, alternatives and modifications are possible within the scope of the appended claims.

This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to make and use the invention. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

What is claimed is:

**1**. A container comprising:

a plurality of side panels each having a pair of side edges;

a plurality of edge brackets configured to join the plurality of side panels to define an open interior of the container, wherein each of the edge brackets comprises:

a first channel member that includes a first receiving channel sized to receive the side edge of a first side panel, wherein the first receiving channel of the first channel member is defined by a back wall, a first side wall and a second side wall, where the first and second side walls are joined to the back wall and extend perpendicular to the back wall; and

a second channel member that includes a second receiving channel sized to receive the side edge of a second side panel, wherein the second receiving channel of the second channel member is defined by a back wall, a side wall and the back wall of the first receiving channel,

wherein the first and second channel members are oriented such that the first and second receiving channels are perpendicular to each other.

**2**. The container of claim **1**, wherein the second side wall of the first channel member is adjacent to and joined to an extended portion of the back wall of the second channel member.

**3**. The container of claim **1** further comprising a top panel and a bottom panel that are each joined to the plurality of side panels that are joined to each other by the plurality of edge brackets.

**4**. The container of claim **2** wherein the back wall of the second channel member extends past the back wall of the first channel member to form the extended portion that is joined to the second side wall of the first channel member.

**5**. The container of claim **1** wherein the plurality of edge brackets are each formed from a metal material.

**6**. The container of claim **1** wherein the first side wall of the first channel member and the side wall of the second channel member each include a plurality of attachment openings.

**7**. An edge bracket for joining two side panels that each have at least a pair of side edges, the edge bracket comprising:

a first channel member that includes a first receiving channel sized to receive the side edge of a first side panel, wherein the first receiving channel of the first channel member is defined by a back wall, a first side wall and a second side wall, where the first and second side walls are joined to the back wall and extend perpendicular to the back wall; and

a second channel member that includes a second receiving channel sized to receive the side edge of a second side panel, wherein the second receiving channel of the second channel member is defined by a back wall, a side wall and the back wall of the first receiving channel,

wherein the first and second channel members are oriented such that the first and second receiving channels are perpendicular to each other.

**8**. The edge bracket of claim **7**, wherein the second side wall of the first channel member is adjacent to and joined to an extended portion of the back wall of the second channel member.

**9**. The edge bracket of claim **7** wherein the back wall of the second channel member extends past the back wall of the first channel member to form the extended portion that is joined to the second side wall of the first channel member.

**10**. The edge bracket of claim **7** wherein the edge bracket is formed from a metal material.

**11**. The edge bracket of claim **7** wherein the first side wall of the first channel member and the side wall of the second channel member each include a plurality of attachment openings.

**12**. The edge bracket of claim **7** wherein a width of the second receiving channel is defined by a distance between the side wall of the second receiving channel and the back wall of the first receiving channel.

**13**. The edge bracket of claim **7** wherein the first side wall of the first channel member extends further from the back wall of the first channel member than the second side wall of the first channel member to define a first side wall extension and the side wall of the second channel member extends further from the back wall of the second channel member than the back wall of the first channel member to define a second side wall extension.

**14**. The edge backet of claim **13** wherein first side wall extension extends at an angle away from the first receiving channel and the second side wall extension extends at an angle away from the second receiving channel.

**15**. The container of claim **1** wherein the first side wall of the first channel member extends further from the back wall of the first channel member than the second side wall of the first channel member to define a first side wall extension and the side wall of the second channel member extends further from the back wall of the second channel member than the back wall of the first channel member to define a second side wall extension.

**16**. The container of claim **15** wherein first side wall extension extends at an angle away from the first receiving channel and the second side wall extension extends at an angle away from the second receiving channel.

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